

TrustedWork

Tunisia

Software Requirements Specification

Platform Architecture & Module Definitions

Version 1.0 | Spring Boot + Angular 18 | 2025

Platform	TrustedWork Tunisia
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1. Project Overview

TrustedWork Tunisia is a next-generation professional freelance marketplace designed specifically for the Tunisian market. The platform addresses three critical challenges in the local economy: youth unemployment, lack of trust in freelance transactions, and the difficulty of finding verified technical talent.

The platform is architected as a microservices system composed of eight independent functional modules. Each module owns its domain, its data, and its business logic. Modules interact exclusively through well-defined REST APIs, ensuring loose coupling and independent deployability.

Problem	Impact	Module Addressing It
Youth unemployment	High	Modules 02, 03, 04, 07
Lack of freelance trust	Critical	Modules 01, 05
Difficulty finding verified talent	High	Modules 02, 03, 04
No structured project execution	Medium	Modules 07, 08
No community for local freelancers	Medium	Modules 04, 06

2. Platform Architecture

The platform follows a microservices architecture where each module is deployed as an independent service with its own database. Inter-module communication uses synchronous REST API calls for request-response interactions. The User Service acts as the shared authentication provider, with all modules validating JWT tokens independently.

Layer	Modules	Responsibility
Trust & Identity	01	User verification and access control
User Value	02, 04	Profile building and skill development
Marketplace	03, 05	Opportunity discovery and contract execution
Execution	07, 08	Team and project management
Engagement	06	Community activation and gamification

Note: Modules share only the user identifier (userId) via JWT token. No module directly accesses another module's database. All cross-module data needs are resolved through REST API calls, ensuring true independence.

3. Technology Stack

Layer	Technology	Purpose
Backend	Spring Boot 3	REST API microservices
Security	Spring Security + JWT	Authentication and authorization
Persistence	JPA / Hibernate + MySQL	Data modeling and storage
Frontend	Angular 18	Single-page application
API Docs	Swagger / OpenAPI 3	API documentation and testing
Build	Maven	Dependency management
Payments	Flouci API, D17 API	Tunisian local payment integration

4. Module Specifications

01

■ Digital Identity & Access Trust

Security gateway

Overview

This module is the entry point of the entire TrustedWork platform. It handles all aspects of user identity, authentication, and access control. It ensures that every user on the platform is a real, verified individual, using advanced biometric and document verification techniques adapted to the Tunisian market.

Data Entities

- User — core account with roles and status
- Session — active login sessions per device
- Device — trusted devices registry
- KYCRequest — identity verification workflow
- AuditLog — immutable security event history
- SuspensionRecord — account restriction history

Core Features

- Multi-role registration (Freelancer, Client, Admin)
- JWT-based authentication with refresh token rotation
- Two-Factor Authentication (2FA) via email/SMS
- Session and device management with revocation
- Account suspension and reactivation workflow
- Full audit trail for all sensitive operations

Advanced Features

#	Feature
1	OCR-based CIN (Tunisian National ID) extraction and validation
2	Liveness detection — real-time selfie vs. CIN photo comparison
3	Automatic Trust Level computation (1–5) post-KYC
4	Device fingerprinting for suspicious login detection
5	Session Risk Score based on geolocation and behavior patterns
6	Automatic account lockout after N failed authentication attempts

AI & Intelligence Components

Component	Description
Liveness Detection Model	Computer vision model comparing selfie to CIN photo to prevent identity spoofing
Anomaly Detection	Detects unusual login patterns (new country, unknown device) and triggers re-authentication
Risk Scoring Engine	Computes a real-time session risk score based on behavioral signals

Key Algorithm

$\text{Trust Level} = (\text{KYC_validated} \times 40) + (\text{2FA_enabled} \times 20) + (\text{Liveness_passed} \times 30) + (\text{Account_age_score} \times 10)$

Technical Stack: Spring Boot 3 | Spring Security | JWT | MySQL | Angular 18

Overview

This module manages the complete professional identity of a freelancer beyond authentication. It provides tools to build a rich, verified portfolio, declare skills with associated levels, and display certifications. It serves as the freelancer's public face on the platform and is fully independent from other modules.

Data Entities

- FreelancerProfile — extended professional profile
- PortfolioItem — project showcase with media
- Skill — declared competency with proficiency level
- Certification — validated credential with expiry
- Endorsement — peer skill validation
- ProfileVisibilitySettings — privacy controls

Core Features

- Rich profile builder with bio, headline, and avatar
- Portfolio management with project descriptions and links
- Skill declaration with Junior / Confirmed / Expert levels
- Certification management with expiry alerts
- Peer endorsement system for skill validation
- Profile visibility and privacy settings

Advanced Features

#	Feature
1	Skill Authenticity Score combining portfolio evidence, exams, and endorsements
2	Portfolio Impact Score measuring visual and technical quality
3	Profile completeness score with improvement suggestions
4	Automatic skill level upgrade based on activity and certifications
5	Regional ranking by Tunisian governorate
6	Career Path Recommendation based on current skill set

AI & Intelligence Components

Component	Description
Portfolio Quality Analyzer	Evaluates portfolio coherence, visual quality, and technical depth automatically
Skill Gap Detector	Identifies missing skills compared to top freelancers in the same domain
Career Path AI	Recommends the optimal next skill to learn based on market demand and current profile

Key Algorithm

$$\text{Skill Authenticity} = (\text{portfolio_evidence} \times 0.40) + (\text{exam_score} \times 0.35) + (\text{peer_endorsements} \times 0.25)$$

Technical Stack: Spring Boot 3 | JPA/Hibernate | MySQL | Angular 18

Overview

This module manages the entire lifecycle of job offers on the platform, from publication by clients to application by freelancers. It goes beyond a simple listing system by incorporating intelligent matching algorithms that connect the right freelancer to the right opportunity based on skills, history, and compatibility scoring.

Data Entities

- JobOffer — mission published by a client
- Application — freelancer application to a job
- MatchScore — computed compatibility between freelancer and job
- SkillVector — normalized skill profile for matching
- CompatibilityReport — detailed match explanation
- OfferFlag — fraud or quality issue signal

Core Features

- Job offer publication with categories, budget, and deadline
- Advanced search and filtering (skill, location, budget, duration)
- Application management for freelancers and clients
- Application status tracking (Pending, Reviewed, Accepted, Rejected)
- Saved offers and application history
- Client dashboard for offer and applicant management

Advanced Features

#	Feature
1	AI-based compatibility scoring between freelancer profile and job requirements
2	Reverse matching — platform proactively suggests jobs to freelancers
3	Skill DNA — normalized competency vector for precise matching
4	Success Prediction Score estimating probability of successful delivery
5	Budget Intelligence detecting underpaid offers vs. market rates
6	Fraud job detection flagging suspicious or scam offers before publication

AI & Intelligence Components

Component	Description
Matching Algorithm	Rule-based and weighted scoring engine computing compatibility between freelancer skills and job requirements
Success Predictor	Estimates contract success probability using historical data and profile features
Fraud Job Classifier	Detects suspicious job offers using NLP pattern analysis on descriptions

Key Algorithm

Match Score = (skills_match x 0.40) + (trust_score x 0.30) + (success_rate x 0.20) + (budget_fit x 0.10)

Technical Stack: Spring Boot 3 | JPA/Hibernate | MySQL | Angular 18

Overview

This module empowers freelancers to continuously improve their skills through a community-driven e-learning platform. Expert freelancers can create and share courses, while learners follow personalized learning paths. The module also manages internal certification exams that validate competencies and are verifiable by employers via QR code.

Data Entities

- Course — learning content with sections and resources
- CourseSection — structured lesson within a course
- Quiz — knowledge assessment attached to a course
- LearningProgress — learner advancement tracking
- Certification — issued credential post-exam
- ExamAttempt — recorded certification exam session
- Instructor — verified course creator profile
- LearningPath — curated sequence of courses

Core Features

- Course creation and publication by community instructors
- Video, article, and quiz-based learning content
- Learning progress tracking with completion percentage
- Internal certification exams with automated grading
- Certification issuance with QR code verification
- Hackathons and technical workshops management

Advanced Features

#	Feature
1	Adaptive exam engine adjusting difficulty based on learner performance
2	Personalized Learning Path AI recommending optimal course sequences
3	Course Relevance Score matching courses to freelancer skill gaps
4	Certification expiry management with renewal reminders
5	Instructor quality scoring based on learner outcomes
6	Regional ranking of certified freelancers by Tunisian governorate

AI & Intelligence Components

Component	Description
Adaptive Exam Engine	Dynamically adjusts question difficulty based on learner response patterns
Learning Path Recommender	Suggests the optimal course sequence based on skill gaps and career goals
Course Quality Scorer	Evaluates course effectiveness using completion rates and learner performance

Key Algorithm

$$\text{Course Relevance} = (\text{skill_gap_match} \times 0.50) + (\text{community_rating} \times 0.30) + (\text{completion_rate} \times 0.20)$$

Technical Stack: Spring Boot 3 | JPA/Hibernate | MySQL | Angular 18

Overview

This module governs the formal working relationship between clients and freelancers. It covers the full lifecycle from contract creation and digital signing to milestone-based payment release via escrow, and handles dispute resolution through an automated arbitration engine. It integrates with Tunisian local payment solutions to ensure accessibility.

Data Entities

- Contract — formal agreement between client and freelancer
- SmartClause — auto-generated contract clause
- Milestone — project delivery checkpoint with payment
- EscrowAccount — locked funds pending milestone validation
- Payment — financial transaction record
- Dispute — raised conflict between parties
- ArbitrationCase — formal dispute resolution process

Core Features

- Digital contract creation with customizable terms
- Electronic signature workflow (E-signature)
- Milestone definition with deadlines and deliverables
- Escrow-based payment locking and release
- Integration with Flouci and D17 (Tunisian wallets)
- Dispute submission and evidence management

Advanced Features

#	Feature
1	Smart Clause Generator automatically drafting contract clauses per mission type
2	Contract Health Score predicting dispute probability before signing
3	Payment Risk Score evaluating client payment reliability
4	Automatic escrow release after 48h of client inactivity
5	Milestone Completion Predictor detecting likely delivery delays
6	Automated arbitration resolving 80% of disputes without human intervention
7	Split payment distribution across team members in agency contracts

AI & Intelligence Components

Component	Description
Contract Health Predictor	Estimates the probability of a dispute based on contract clarity, budget, and party profiles
Arbitration Engine	Rule-based decision system resolving standard disputes automatically using predefined criteria
Delay Predictor	Anticipates milestone delivery delays using historical delivery patterns

Key Algorithm

$$\text{Contract Health} = (\text{client_trust} \times 0.30) + (\text{freelancer_trust} \times 0.30) + (\text{contract_clarity} \times 0.20) + (\text{budget_fairness} \times 0.20)$$

Technical Stack: Spring Boot 3 | JPA/Hibernate | MySQL | Angular 18 | Flouci API | D17 API

Overview

This module drives platform engagement through two complementary mechanics: a structured events system (hackathons, meetups, webinars) and a gamification engine rewarding community participation. It operates independently, tracking engagement-specific interactions without depending on contract or payment data.

Data Entities

- Event — hackathon, meetup, or webinar
- EventRegistration — participant enrollment
- GrowthProfile — XP and level tracking per user
- Badge — achievement awarded for specific actions
- UserBadge — badge assignment record
- Streak — consecutive participation tracking
- Leaderboard — ranked engagement scoreboard
- Challenge — community-defined engagement task

Core Features

- Event creation and management (online and in-person)
- Event registration with capacity management
- Geolocation-based event discovery by Tunisian city
- XP points system rewarding community participation
- Badge system with unlock conditions and rarity levels
- Regional leaderboard ranked by governorate

Advanced Features

#	Feature
1	Streak system rewarding consecutive event participation
2	Challenge engine with community-defined objectives
3	Engagement Score computing overall platform activity level
4	Churn Prediction detecting users at risk of disengagement
5	Smart Event Matching recommending events by skills and location
6	Influence Score measuring a user's community impact

AI & Intelligence Components

Component	Description
Churn Prediction Model	Identifies users showing early signs of disengagement and triggers re-activation campaigns
Smart Event Recommender	Matches users to relevant events based on skill profile, location, and past participation
Engagement Scorer	Computes a composite engagement score from multiple community interaction signals

Key Algorithm

Engagement Score = (event_participation x 0.30) + (challenge_completion x 0.25) + (streak_length x 0.25) + (community_influence x 0.20)

Technical Stack: Spring Boot 3 | JPA/Hibernate | MySQL | Angular 18

Overview

This module introduces a unique concept to the Tunisian freelance market: the ability to form and manage freelance agencies. Agencies group multiple freelancers under a shared brand, allowing them to bid on larger contracts collectively. The module manages team structure, task assignment, internal collaboration, and agency-level performance tracking.

Data Entities

- Agency — registered freelance collective with branding
- AgencyMember — membership record with role
- TeamProject — project assigned to an agency
- Task — granular work item within a project
- TaskAssignment — member-to-task allocation
- CollaborationLog — internal communication record
- AgencyPerformanceScore — computed agency reliability metric

Core Features

- Agency creation and profile management
- Member invitation and role management (Lead, Member, Observer)
- Internal task board with assignment and status tracking
- Team project dashboard with progress visualization
- Internal messaging and collaboration log
- Agency public profile visible to clients

Advanced Features

#	Feature
1	Agency Performance Score based on delivery history and client ratings
2	Workload balancing recommendation across agency members
3	Skill coverage analysis ensuring the agency can fulfill a contract
4	Automatic task distribution based on member availability and skills
5	Agency tier system (Starter, Verified, Premium)
6	Conflict of interest detection in task assignments

AI & Intelligence Components

Component	Description
Workload Balancer	Recommends optimal task distribution across agency members based on skills and current load
Skill Coverage Analyzer	Verifies that an agency collectively holds all required skills before bidding on a contract
Performance Predictor	Estimates agency success probability for a given contract type

Key Algorithm

Agency Score = (delivery_rate × 0.35) + (client_satisfaction × 0.30) + (response_time × 0.20) + (member_retention × 0.15)

Technical Stack:

Spring Boot 3 | JPA/Hibernate | MySQL | Angular 18

Overview

This module provides a full project management suite inside the platform, enabling freelancers and clients to collaboratively manage project execution. It goes beyond basic task management by incorporating AI-powered delivery intelligence that proactively detects delays, predicts bottlenecks, and provides actionable insights to keep projects on track.

Data Entities

- Project — top-level engagement with scope and timeline
- Task — work item with status, priority, and assignee
- SubTask — granular decomposition of a task
- Deliverable — formal output submitted for client review
- GanttEntry — time-based task scheduling record
- KanbanBoard — visual workflow state machine
- ProgressReport — automated project health snapshot
- DeliveryRiskSignal — AI-generated delay warning

Core Features

- Project creation with scope, timeline, and budget definition
- Task and subtask management with priority and status
- Kanban board with drag-and-drop workflow management
- Gantt chart visualization for timeline planning
- Deliverable submission and client review workflow
- Automated progress reports with completion percentage

Advanced Features

#	Feature
1	AI delay detection identifying tasks at risk before deadlines are missed
2	Delivery Risk Score computed per project in real time
3	Bottleneck analyzer identifying tasks blocking overall progress
4	Smart deadline recommendation based on task complexity and assignee history
5	Automatic progress report generation with key insights
6	Historical delivery pattern analysis per freelancer

AI & Intelligence Components

Component	Description
Delay Detection Engine	Analyzes task velocity, dependency chains, and historical patterns to flag at-risk deliverables early
Bottleneck Analyzer	Identifies tasks creating downstream delays and suggests re-prioritization
Smart Deadline Recommender	Suggests realistic deadlines based on task complexity and assignee delivery history

Key Algorithm

$$\text{Delivery Risk} = (\text{days_remaining} / \text{estimated_duration} \times 0.35) + (\text{task_completion_rate} \times 0.35) + (\text{assignee_history_score} \times 0.30)$$

Technical Stack: Spring Boot 3 | JPA/Hibernate | MySQL | Angular 18

5. Integration Strategy

All modules communicate exclusively through REST APIs. No module directly accesses another module's database. The only shared element across modules is the `userId` extracted from the validated JWT token. This design ensures that each module can be developed, tested, and deployed independently.

Interaction	Type	Description
Authentication	JWT Token	All modules validate tokens issued by Module 01
Profile data	REST GET	Module 03 reads basic profile from Module 02 API
Contract creation	REST POST	Module 05 is triggered after offer acceptance in Module 03
Certification badge	REST GET	Module 03 reads certification status from Module 04
Agency contract	REST	Module 07 interacts with Module 05 for team contracts
Project from contract	REST POST	Module 08 creates project after contract signing in Module 05

Important: These interactions represent loose coupling only. Each module functions independently. If any module is unavailable, other modules continue to operate normally using cached or default values.

6. Summary

#	Module	Domain	AI Features	Key Innovation
01 ■	Identity & Access	Security	3	OCR CIN + Liveness Detection
02 ■	Freelancer Profile	User Value	3	Skill Authenticity Score
03 ■	Job Board & Matching	Marketplace	3	DNA-based Smart Matching
04 ■	Learning & Certification	Education	3	Community-driven E-Learning
05 ■■	Contracting & Escrow	Finance	3	Auto Arbitration + Flouci
06 ■	Events & Gamification	Engagement	3	Churn Prediction Engine
07 ■	Agency Management	Collaboration	3	Freelance Agency System
08 ■	Project Management	Execution	3	AI Delay Detection

TrustedWork Tunisia is designed as a scalable, modular, and maintainable platform. Each module represents a complete business domain with independent value, advanced business logic beyond simple CRUD operations, and integrated AI capabilities. The platform addresses real market needs in Tunisia and is built on industry-standard technologies suitable for production deployment.